IGF-1 LR3

Insulin-Like Growth Factor 1 (Long R3) | 1mg x 10 Vials

Muscle Hyperplasia & Nutrient Partitioning -- Post-Workout Injection

Purity: 99.8% | Endotoxin-Free | Research Use Only

IGF-1 LR3 (Long R3 Insulin-like Growth Factor-1) is a synthetic analogue of endogenous IGF-1, with an arginine substitution at position 3 and a 13-amino acid N-terminal extension. These modifications dramatically increase half-life from ~10 minutes (native IGF-1) to 20-30 hours, and reduce binding to IGF binding proteins (IGFBPs) by >1000-fold, making a significantly higher proportion of the peptide biologically active.

CRITICAL: IGF-1 LR3 can cause hypoglycaemia. Always have fast-acting carbohydrates (glucose tablets, juice) available during and after injection. NEVER inject fasted. Inject immediately post-workout when muscles are insulin-sensitive.

Reconstitution:

+ 1 mL BAC Water
= 1 mg/mL
= 1,000 mcg/mL
100 doses at 10mcg
20 doses at 50mcg

Dose Range:

20-40mcg Beginner
50-60mcg Standard
80-100mcg Advanced

Post-workout ONLY

1mg / 1mL BAC Water -> 1,000 mcg/mL

Parameter	Specification
Compound	IGF-1 LR3 -- Long R3 analogue of Insulin-like Growth Factor 1
Molecular Weight	~9,200 Da (83 amino acids)
Half-Life	20-30 hours (vs ~10 min for native IGF-1) -- due to reduced IGFBP binding
Vial Size	1 mg per vial | 10 vials per kit | 10 mg total
Reconstitution	1 mL BAC Water -> 1 mg/mL = 1,000 mcg/mL
Doses per Vial	50 doses at 20mcg | 20 doses at 50mcg | 10 doses at 100mcg
Standard Dose	20-100 mcg per injection -- start at 20-40 mcg for first cycle
Frequency	Once daily post-workout on training days -- NOT on rest days
Timing (CRITICAL)	Inject within 30 minutes of completing workout -- before muscle insulin sensitivity window closes
Route	SubQ near trained muscle (preferred) or IM into trained muscle
Cycle	4-6 weeks ON, min 4 weeks OFF | Storage: 2-8 deg C unreconstituted; reconstituted use within 21 days

MECHANISM OF ACTION & RESEARCH BENEFITS

IGF-1 LR3 binds to IGF-1 receptors (IGF-1R) on target cells, activating the PI3K/Akt/mTOR and MAPK/ERK signalling cascades. The Long R3 modification minimises sequestration by circulating IGFBPs, keeping the peptide free and active in tissue for up to 30 hours. Post-workout injection exploits the heightened IGF-1R sensitivity and nutrient partitioning in recently exercised muscle.

Pathway	Mechanism	Research Application
PI3K/Akt/ mTOR	IGF-1R activation triggers PI3K; phosphorylates Akt; activates mTORC1 and mTORC2; drives protein synthesis via S6K1 and 4E-BP1; inhibits protein breakdown via FOXO transcription factors	Muscle protein synthesis, lean mass accrual, anti-catabolic protection
Muscle Hyperplasia	Unlike IGF-1 or HGH which primarily cause hypertrophy, IGF-1 LR3 stimulates satellite cell proliferation and differentiation -- increasing the number of muscle cells (hyperplasia), not just their size	Unique muscle cell proliferation effect; permanent new muscle fibre research
Nutrient Partitioning	Enhances GLUT4 translocation in muscle; increases amino acid and glucose uptake; redirects nutrients preferentially into muscle tissue over adipose; insulin-sensitising effect in muscle	Improved nutrient uptake by trained muscle; body composition research
Anti-Lipogenic Action	Reduced IGFBP binding means less systemic IGF-1 activity in fat tissue; preferential action on exercised muscle; some lipolytic signalling via insulin-pathway crosstalk	Fat loss alongside muscle growth; body recomposition potential
Hypoglycaemia Risk	IGF-1 receptors are structurally similar to insulin receptors; IGF-1 LR3 has partial insulin-receptor activity; lowers blood glucose -- risk is dose and timing dependent	MUST inject post-workout with carbohydrates available; monitor blood glucose; have glucose source ready

Key Research Benefits

Benefit	Evidence & Research Context
Muscle Hyperplasia (Unique)	IGF-1 LR3 is one of the few peptides shown to stimulate satellite cell proliferation -- potentially increasing muscle fibre number, not just size. This effect is distinct from HGH or standard IGF-1.
Extended Active Window	The 20-30 hour half-life means once-daily dosing provides sustained receptor engagement, unlike native IGF-1 which clears within minutes.
Synergy with Training	Post-workout injection into exercised muscle maximises uptake in hypersensitive tissue, improving nutrient partitioning and satellite cell response.
Anti-Catabolic Protection	mTOR activation and FOXO inhibition reduce muscle protein breakdown, protecting lean mass during caloric deficit or intense training phases.

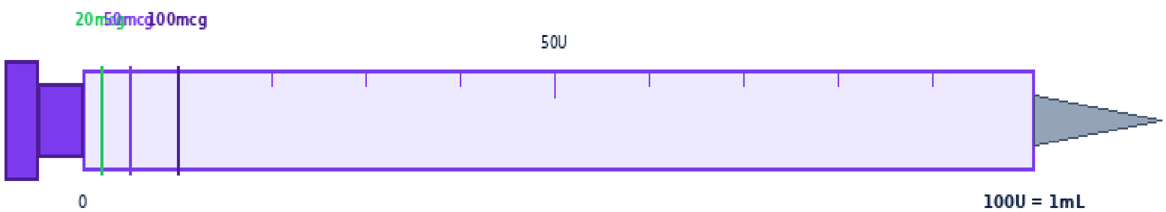
RECONSTITUTION PROTOCOL & DOSE GUIDE

IGF-1 LR3 is sensitive to temperature and agitation. Chill the vial at 4 deg C before reconstitution — cold temperature slows enzymatic degradation and preserves peptide integrity during reconstitution. Use 1 mL BAC Water per vial, yielding 1,000 mcg/mL. Never shake. Roll gently. Use within 21 days of reconstitution.

Step 1: Chill vial at 4C before reconstituting -- reduces IGF-1 degradation.

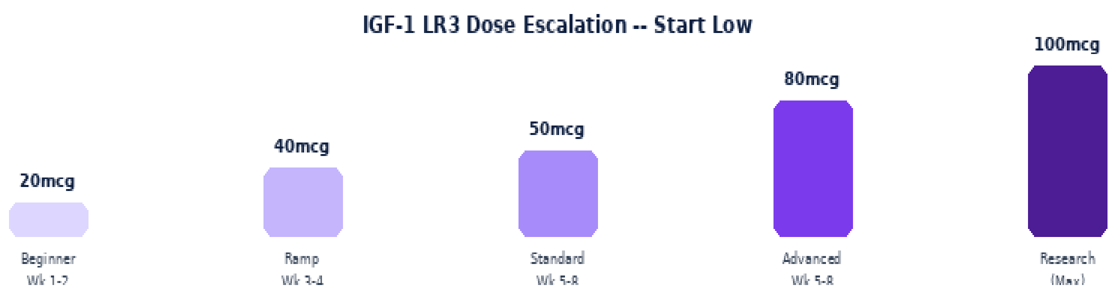
Step	Action	Detail
1	Chill vial at 4 deg C	Place the dry IGF-1 LR3 vial in the fridge at 4 deg C for 30 minutes before reconstitution. Also chill the BAC Water. Cold temperature minimises peptide degradation during dissolution.
2	Clean stoppers	Wipe rubber stoppers of BOTH vials with alcohol swab. Air-dry 10-15 seconds.
3	Draw 1 mL BAC Water	Draw exactly 1 mL BAC Water into a clean insulin syringe. Remove all air bubbles.
4	Inject BAC Water into IGF-1 LR3 vial	Direct stream against the GLASS WALL -- never onto the powder. Inject slowly. IGF-1 LR3 begins dissolving on contact.
5	Dissolve gently	Roll between palms 15-30 seconds. Do NOT shake or vortex. Solution should be clear and colourless within 60 seconds.
6	Label and refrigerate	Write reconstitution date. Store at 2-8 deg C. Use within 21 days. Never freeze. Keep away from light.

Dose Calculation Guide



Insulin Syringe 1mL/100U | 29-31G needle | At 1,000 mcg/mL: 1U = 10 mcg

At 1,000 mcg/mL: 1 Unit on an insulin syringe = 10 mcg | **Formula:** Volume (mL) = Desired Dose (mcg) / 1,000 | **Units:** Dose (mcg) / 10 = Units on syringe



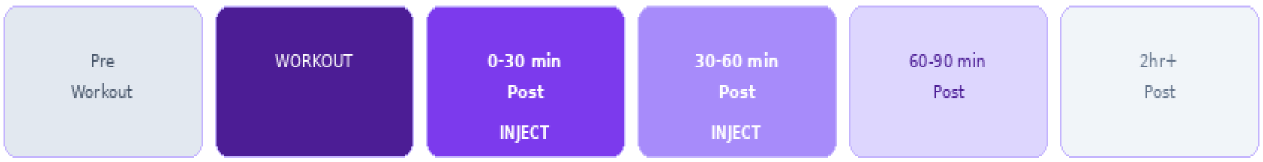
Vial: 1mg x 10 -- Add 1 mL BAC Water → Concentration: 1,000 mcg/mL

Dose (mcg)	20 mcg	40 mcg	50 mcg	60 mcg	80 mcg	100 mcg
Volume (mL)	0.0200	0.0400	0.0500	0.0600	0.0800	0.1000
Syringe Units	2.0U	4.0U	5.0U	6.0U	8.0U	10.0U
Doses / Vial	50	25	20	16	12	10

INJECTION TECHNIQUE, SITES & TIMING

Post-workout timing is the most critical variable with IGF-1 LR3. Inject within 30 minutes of completing your training session, near or into the muscle groups you just trained. This exploits peak GLUT4 expression, increased IGF-1 receptor density, and enhanced blood flow to deliver maximum satellite cell stimulation and nutrient partitioning.

Post-Workout Injection Timing Window



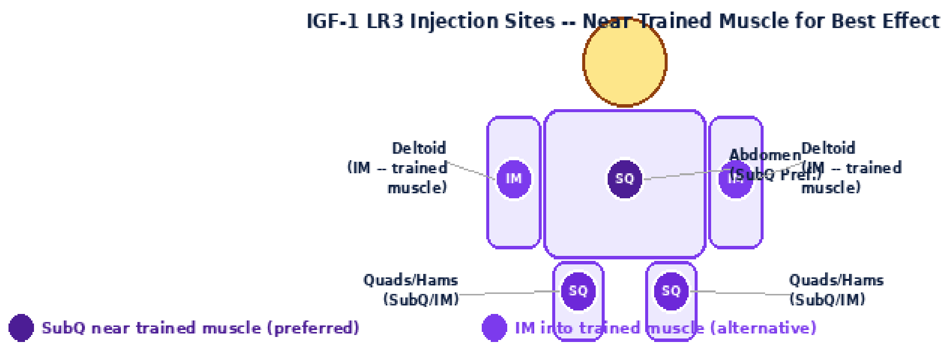
Best: 0-30 min post-workout. Acceptable: up to 60 min. Avoid 90min+ delay.

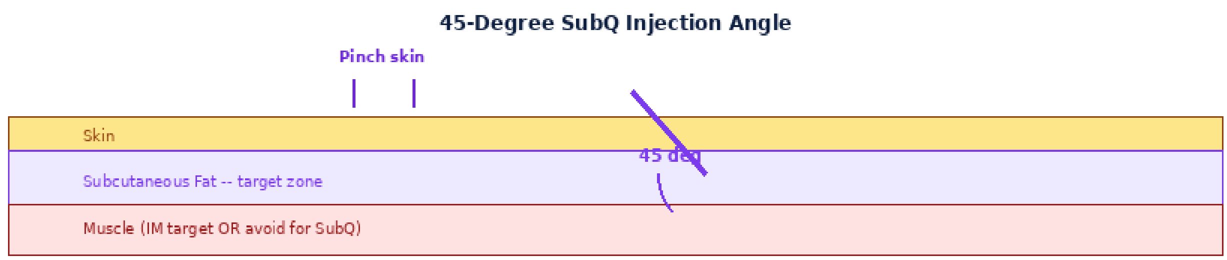
Step	Action	Detail
1	Prepare carbohydrates	Have 20-40g fast-acting carbohydrates ready BEFORE injecting (glucose tablets, dextrose, juice). IGF-1 LR3 can lower blood glucose. Consume them within 15 minutes of injection.
2	Wash hands	Wash 20 seconds with soap and warm water. Dry completely.
3	Select needle	SubQ: 29-31G x 5/16 inch insulin syringe. IM: 23-25G x 1 inch. Volumes are very small -- use 0.3 mL syringe for doses under 50 mcg for more precise measurement.
4	Draw dose precisely	Draw exact volume -- verify twice. At 1,000 mcg/mL doses are very small (e.g. 50 mcg = 5 Units). Expel all air bubbles.
5	Select site near trained muscle	SubQ into abdominal or limb fat near the trained muscle group. E.g. after leg day: inject near quads. After chest/arms: abdominal SubQ.
6	Clean site and inject	Wipe with alcohol swab. Dry 10 sec. Pinch skin. 45-degree angle. Release pinch. Inject slowly over 5-10 seconds.
7	Withdraw and consume carbs	Remove needle. Press with dry swab 5 sec. Do NOT rub. Consume your post-workout carbohydrates immediately if not already done.
8	Monitor blood glucose	Especially in first few sessions, monitor for hypoglycaemia symptoms: shakiness, sweating, confusion, rapid heartbeat. Have glucose source immediately available.
9	Rotate and dispose	Rotate injection site. Needle into sharps container immediately.

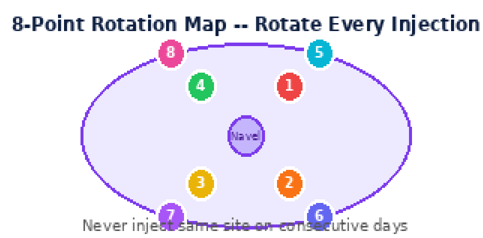
INJECTION SITES & ROTATION

Inject **near or into the muscle group trained that session** for maximum satellite cell stimulation and nutrient partitioning. SubQ into abdominal fat is the most convenient universal site. IM into the trained muscle is preferred by advanced researchers. Rotate systematically to prevent site fatigue.

Site	Route	Needle	Best After	Notes
Abdomen (SubQ)	SubQ	29-31G 5/16 in	Any workout	Most convenient; good systemic delivery; easy to rotate 8 points.
Quads (SubQ/IM)	SubQ or IM	29-31G (SQ) 23-25G (IM)	Leg day	Inject near trained quad for localised muscle effect.
Deltoid (IM)	IM	23-25G 1 inch	Push/arm day	Into deltoid after shoulder or arm training.







Research Dosing Protocols

Protocol	Dose	Timing	Cycle	Key Notes
Beginner	20-40 mcg	0-30 min post-workout	4 weeks on 4 weeks off	Start at 20mcg. Training days only. Have carbs ready.
Intermediate	50-60 mcg	0-30 min post-workout	4-6 weeks on 4 weeks off	Standard research dose. Training days only. Monitor glucose.
Advanced	80-100 mcg	0-30 min post-workout	4-6 weeks on 6 weeks off	Higher hypoglycaemia risk. Glucose monitoring essential.
Off Cycle (Mandatory)	0 mcg	N/A	Min 4 weeks	Essential -- prevents IGF-1R desensitisation and maintains efficacy.

Side Effects

Side Effect	Likelihood	Management
Hypoglycaemia	Common	Most significant risk. Always inject post-workout with carbs available. Symptoms: shakiness, sweating, confusion, rapid heartbeat, pallor. Treat immediately with fast glucose (15-20g dextrose/juice).
Lethargy / fatigue post-injection	Common	Blood glucose drop. Consume carbohydrates immediately post-injection. Reduces with experience and correct timing.
Nausea	Occasional	Often glucose-related. Ensure adequate post-workout carbohydrate intake.
Joint / tendon discomfort	Occasional	Water retention in joint spaces. Common at higher doses. Reduce dose if persistent.
Jaw / facial tingling	Occasional	Growth-stimulating effect on facial tissues. Reduce dose. Usually resolves on cessation.
IGF-1R desensitisation	With long cycles	Continuous use >6 weeks causes receptor downregulation and loss of response. Mandatory 4-week minimum off-cycle.
Cell proliferation (cancer risk)	Long-term concern	IGF-1 signalling promotes cell growth. Contraindicated in any cancer history. Short cycles with breaks minimise risk.

STOP & SEEK MEDICAL ADVICE: Severe hypoglycaemia: loss of consciousness, seizure, inability to self-treat -- call emergency services

STOP & SEEK MEDICAL ADVICE: Any known or suspected cancer or pre-cancerous condition

STOP & SEEK MEDICAL ADVICE: Severe or rapidly worsening headache, visual disturbance, or neurological symptoms

STOP & SEEK MEDICAL ADVICE: Anaphylaxis: hives, facial swelling, difficulty breathing

Absolute Contraindications: Any history of cancer or pre-cancerous conditions (IGF-1 stimulates cell proliferation), active hypoglycaemia-prone conditions without glucose monitoring, diabetes requiring insulin (hypoglycaemia risk amplified), pregnancy or lactation. Not for use in individuals under 18.

Storage Guidelines

State	Temperature	Duration	Notes
Unreconstituted powder	2-8 deg C	Per expiry	Keep upright. Away from light. Do not freeze.
Reconstituted solution	2-8 deg C	21 days	Label with date. Use within 21 days (IGF-1 LR3 degrades faster than most peptides). Inspect before each use.
BAC Water (opened)	2-8 deg C	28 days	Label opening date.
Transport	Below 20 deg C	Max 48 hr	Insulated bag with ice packs. IGF-1 LR3 is temperature-sensitive.

Golden Rules -- IGF-1 LR3

1. ALWAYS have fast-acting carbohydrates ready before injecting -- hypoglycaemia risk is real.
2. Inject ONLY post-workout, within 30 minutes of completing training -- timing is critical.
3. NEVER inject on rest days -- receptor desensitisation accelerates with unnecessary dosing.
4. Chill the vial at 4 deg C before reconstitution -- preserves peptide integrity.
5. Use 1 mL BAC Water per vial. At 1,000 mcg/mL: 1 syringe unit = 10 mcg.
6. Use a 0.3 mL syringe for doses under 50 mcg -- small volumes require precise measurement.
7. Inject near or into the muscle group trained that session for maximum localised effect.
8. Cycle strictly: 4-6 weeks ON followed by minimum 4 weeks OFF. No exceptions.
9. Store reconstituted IGF-1 LR3 at 2-8 deg C. Use within 21 days -- not 28 like other peptides.
10. Absolutely contraindicated in any cancer history -- IGF-1 signalling stimulates cell proliferation.